LEADER & ACHIEVER COURSE

PHASE : MLA,M,N,W,X,MAZC,D,E,O,P,U/V,MAAK,U/V,Y,Z,MAPB,MSP (SPARK PRO)

TARGET : PRE-MEDICAL 2026

Test Type : MAJOR

Test Pattern : NEET (UG)

TEST DATE : 30-03-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	1	3	4	3	2	1	3	3	1	3	1	1	2	2	3	1	2	2	1	3	2	3	1	1	1	4	3	2	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	2	1	3	2	3	3	3	4	3	4	2	1	1	3	4	4	4	3	1	4	4	3	4	1	1	2	1	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	1	1	4	4	2	3	3	2	4	4	3	1	3	1	1	4	1	3	4	4	3	2	2	4	4	4	1	2	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	4	3	3	1	1	1	3	4	1	3	1	3	1	2	4	1	2	2	4	3	4	3	2	2	4	2	2	3	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	2	3	2	2	3	2	2	3	3	3	1	3	2	1	1	2	2	1	3	1	1	2	4	2	4	4	1	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	4	3	3	1	3	3	1	3	1	4	1	3	1	2	4	1	2	4	4	1	2	1	4	4	1	3	2	2	2

HINT – SHEET

1. Ans (3)

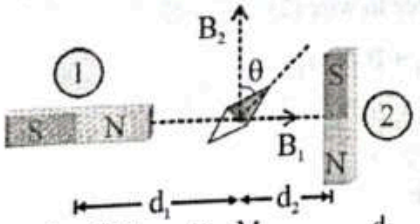
Magnet A produces magnetic field along the axial (end-on) direction at P, $B_1 = \frac{2M}{4\pi d_1^3}$

Magnet B produces magnetic field along the equatorial (broadside) direction at P, $B_2 = \frac{M}{4\pi d_2^3}$

The needle deflects by angle θ, so by tangent law,

In equilibrium $\tan\theta = \frac{B_1}{B_2}$

$B_1 = B_2 \tan\theta$



$$\Rightarrow \frac{\mu_0}{4\pi}\cdot\frac{2M}{d_1^3} = \frac{\mu_0}{4\pi}\cdot\frac{M}{d_2^3}\tan\theta \Rightarrow \frac{d_1}{d_2} = (2\cot\theta)^{1/3}$$

2. Ans (1)

$F_{AB} = F_{BC}$

$$\frac{\mu_0(i)}{2\pi}\frac{(2i)}{d_1} = \frac{\mu_0}{2\pi}\frac{(2i)(3i)}{d_2}$$

$$\therefore \frac{d_1}{d_2} = \frac{1}{3}$$

3. Ans (3)

$F = Bi\,\ell_{eff}$

$F = Bi \times 2L$

4. Ans (4)

Remains constant whatever may be velocity of projection of given charged particle

$\omega = \frac{qB}{m}$

Scanned with OKEN Scanner

5. **Ans (3)**

$$B_{in} = \frac{\mu_0 ir}{2\pi R^2} = \frac{\mu_0 ia}{2\pi \times a^2 \times 2} = \frac{\mu_0 i}{4\pi a}$$

$$B_{out} = \frac{\mu_0 i}{2\pi R} = \frac{\mu_0 i}{2\pi \times 2a} = \frac{\mu_0 i}{4\pi a}$$

$$\frac{B_{in}}{B_{out}} = 1$$

6. **Ans (2)**

$W_1 = MB(\cos 0° - \cos 90°) = MB$

$W_2 = MB(\cos 0° - \cos 60°) = \frac{MB}{2}$

$\therefore W_1 = nW_2$

$MB = n\frac{MB}{2}$

$n = 2$

7. **Ans (1)**

$M = NIA$

$N = 1$

For ABCD

$\vec{M}_1 = abI\ \hat{K}.$

For DEFA

$\vec{M}_2 = abI\ \hat{j}$

$\vec{M} = \vec{M}_1 + \vec{M}_2$

$= ab\,I\left(\hat{k} + \hat{j}\right)$

$\Rightarrow = ab\,I\sqrt{2}\left(\frac{\hat{j}}{\sqrt{2}} + \frac{\hat{k}}{\sqrt{2}}\right)$

8. **Ans (3)**

$\vec{F} = I\left(\vec{\ell} \times \vec{B}\right) = 4\left(8\hat{i} \times 5\hat{k}\right) = -160\,\hat{j}\ N$

9. **Ans (3)**

$B_x = B_0 / (1 + x^2 / R^2)^{3/2}$

Given, $B_x = B_0 / 64$

So,

$B_0 / (1 + x^2 / R^2)^{3/2} = B_0 / 64$

Cancelling B_0 on both sides,

$(1 + x^2 / R^2)^{3/2} = 64$

Taking power 2/3 on both sides,

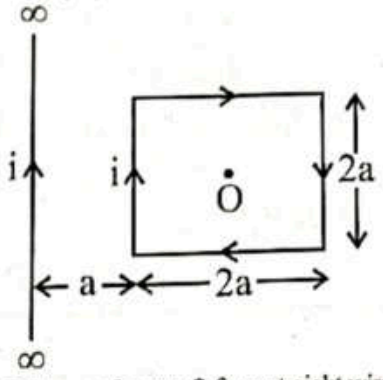
$1 + x^2 / R^2 = 16$

So,

$x^2 / R^2 = 15$

$x = \sqrt{15}\,R$

10. **Ans (1)**



Distance of centre O from straight wire = a + a = 2a

Magnetic field at O due to infinite wire

$$B_{wire} = \frac{\mu_0 i}{2\pi \times 2a} = \frac{\mu_0 i}{4\pi a}$$

For square loop, distance of centre from each side = a

Magnetic field due to one side at centre

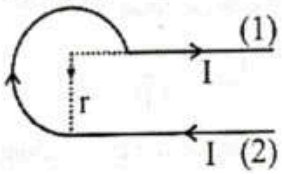
$\theta_1 = \theta_2 = 45°$

$$B_{square} = 4 \times \frac{\mu_0 i}{4\pi a}(\sin 45° + \sin 45°) = \sqrt{2}\frac{\mu_0 i}{\pi a}$$

Direction of magnetic field due to square loop and straight wire at O is same. Net magnetic field $B_{net} = B_{wire} + B_{square}$

$$B_{net} = \frac{\mu_0 i}{4\pi a} + \sqrt{2}\frac{\mu_0 i}{\pi a} = \frac{\mu_0 i}{2\pi a}\left(\frac{1}{2} + 2\sqrt{2}\right)$$

11. **Ans (3)**



B due to Arc

$$B_1 = \frac{\mu_0 I\theta}{4\pi r} = \frac{\mu_0 I}{4\pi r} \times \frac{3\pi}{2}$$

$$= \frac{3\mu_0 I}{8r} \otimes$$

B_2 = B due to wire (1) = 0

$$B_3 = \text{B due to wire (2)} = \frac{\mu_0 I}{4\pi r} \otimes$$

$B_{net} = B_1 + B_2 + B_3$

$$= \frac{\mu_0 I}{4\pi r}\left(\frac{3\pi}{2} + 1\right)$$

12. **Ans (1)**

$$B = \frac{\mu_0 i}{2\pi r} = \frac{2 \times 10^{-7} \times 50}{2.5}$$

$= 4 \times 10^{-6}\ T$

Scanned with OKEN Scanner

13. Ans (1)

Let r be the radius of new arc $\widehat{AB}$

$\therefore\ r(2\pi-\theta)=L$ or $r=\dfrac{L}{(2\pi-\theta)}$

$\therefore$ Magnetic field at centre O is

$$B=\frac{\mu_0 i}{2r}\left(\frac{2\pi-\theta}{2\pi}\right)=\frac{\mu_0\pi i}{L}\left(\frac{2\pi-\theta}{2\pi}\right)^2$$

14. Ans (2)

$\oint \vec{B}.\vec{dl}=\mu_0 i$

$=\mu_0(I_6+I_5+I_4-I_3-I_2-I_1)$

$=\mu_0(4+5+1-3-2-1)$

$=4\,\mu_0$ Wb/m.

15. Ans (2)

Soft iron Low susceptibility

low coercivity

Steel $\Rightarrow$ High $\Rightarrow$ Retentivity

High $\Rightarrow$ Coercivity

16. Ans (3)

Diamagnet $\longrightarrow$ Copper

Paramagnet $\longrightarrow$ Aluminium

Soft ferromagnet $\longrightarrow$ Gadolinium

Hard ferromagnet $\longrightarrow$ Alnico

17. Ans (1)

Magnetising field,

$H=n\times i$

$H=1000\times 2=2\times 10^3$ A m^{-1}

Relative permeability, So,

$\mu_r=1+\chi$ $\qquad$ $\chi=\mu_r-1=400-1=399$

Intensity of magnetization,

$I=\chi\times H$

$I=399\times 2\times 10^3$

$I=798\times 10^3$ A m^{-1}

18. Ans (2)

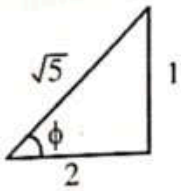
$m=\left(\dfrac{q}{2m}\right)L$

19. Ans (2)

$Z=\sqrt{R^2+(X_L-X_C)^2}$

$Z=\sqrt{R^2+\dfrac{R^2}{4}}=\dfrac{\sqrt{5}R}{2}$

$\cos\phi=\dfrac{R}{Z}=\dfrac{R\times 2}{\sqrt{5}R}=\dfrac{2}{\sqrt{5}}$



$\tan\phi=\dfrac{1}{2}$

$\therefore\ \phi=\tan^{-1}\left(\dfrac{1}{2}\right)$

20. Ans (1)

$V_{rms}=\sqrt{\bar{V}^2}\Rightarrow V_{rms}^2=\bar{V}^2=\dfrac{\int_{t_1}^{t_2}V^2dt}{t_2-t_1}$

$$\therefore\ V_{rms}^2=\frac{\int_0^1\left(V_0t^{3/2}\right)^2dt}{1-0}=V_0^2\int_0^1 t^3dt$$

$$=V_0^2\left[\frac{t^4}{4}\right]_0^1=V_0^2\left[\frac{1}{4}-\frac{0}{4}\right]=\frac{V_0^2}{4}$$

therefore, $V_{rms}=\dfrac{V_0}{2}$

21. Ans (3)

$P=V_{rms}I_{rms}\cos\phi$

$=\dfrac{200\sqrt{2}}{\sqrt{2}}\times\dfrac{2}{\sqrt{2}}\cos 30^\circ$

$=200\sqrt{2}\times\dfrac{\sqrt{3}}{2}=100\sqrt{6}$ watt

22. Ans (2)

at $\phi=\omega t=0$ $\longrightarrow$ $\begin{cases}E=0\\ I=-I_0\end{cases}$

at $\phi=\omega t=\dfrac{\pi}{2}$ $\longrightarrow$ $\begin{cases}E=E_0\\ I=0\end{cases}$

So, emf E leads current I by phase $\dfrac{\pi}{2}$

23. Ans (3)

$Q=\dfrac{1}{R}\sqrt{\dfrac{L}{C}}$

Q $\longrightarrow$ High for best tunning

Scanned with OKEN Scanner

24. Ans (1)

When a bulb and a capacitor are connected in series to an ac source, then on increasing the frequency the current in the circuit is increased, because the impedance of the circuit is decreased. So the bulb will give more intense light.

25. Ans (1)

In csae of Resonance

$$\omega_r = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{8\times10^{-3}\times20\times10^{-6}}}$$

$= 2500$ rad/s

$$i = \frac{V}{R} = \frac{220}{44}A = 5A$$

peak value $= \sqrt{2}\, i_{rms} = 5\sqrt{2}\,A$

26. Ans (1)

Downward magnetic field increases with time so to reduce it induced electric field line will be in anti clock wise direction.

27. Ans (4)

$\phi = BA$

$MI = BA$ here $(x >> R)$

$$M = \frac{BA}{I} = \frac{\mu_0 I R^2 \times \pi r^2}{2x^3 \times I}$$

28. Ans (3)

$$e = -L\frac{di}{dt} = -6\times10^{-3}\left(-\frac{(7-5)}{(5-2)\times10^{-3}}\right) = 4V$$

29. Ans (2)

$$\oint \overrightarrow{E_{ind}}\cdot\overrightarrow{d\ell} = -\frac{d\phi}{dt}$$

$$E_{ind}\cdot(2\pi r) = \frac{AdB}{dt}$$

$$E_{ind} = \frac{\pi a^2}{2\pi r}\frac{dB}{dt}$$

$$E_{ind} = \frac{a^2}{2r}\frac{dB}{dt}$$

$$E_{ind} \propto \frac{1}{r}$$

30. Ans (1)

Motional emf induced in the connector $\varepsilon = vB\ell = (2)(2)(1) = 4V$

This acts as a cell of emf 4V and internal resistance 2Ω

6Ω and 3Ω resistors are in parallel.

$$\therefore \frac{1}{R_p} = \frac{1}{6}+\frac{1}{3} = \frac{1+2}{6} = \frac{3}{6} = \frac{1}{2}\ ; R_P = 2\Omega$$

$\therefore$ Current through the connector,

$$I = \frac{\varepsilon}{R_p + r} = \frac{4}{2+2} = 1A$$

Magnetic force on the connector $F = BI\ell = 2(1)\,(1) = 2N$

31. Ans (3)

The manner in which the two coils are oriented, determines the coefficient of coupling between them.

$M = K\sqrt{L_1 L_2}$

When the two coils are wound on each other, the coefficient of coupling is maximum and hence mutual inductance between the coil is maximum.

32. Ans (2)

Due to Lenz law

33. Ans (1)

According to Lenz's law, induced current will be developed in both the loops in opposite direction. Therefore, the net current in each loop will decrease.

34. Ans (3)

Induced voltage $|v| = -L\frac{di}{dt}$.

$i = 3t\sin t$

$L = 1H$

$$V = -1\times\frac{di}{dt}(3t\sin t)$$

$V = -3\sin t - 3t\cos t$

35. Ans (2)

Rate of growth of current in L–R circuit.

$$\frac{dI}{dt} = \frac{E}{L}(e^{-t/\lambda})$$

At $t = 0$

$$\frac{dI}{dt} = \frac{E}{L} \Rightarrow 4 = \frac{E}{20}$$

$E = 80$ volt

Scanned with OKEN Scanner

36. Ans (3)

According to lenz law induce current, oppose the couses of change in flux

37. Ans (3)

Displacement current is equal to conduction current whether the source is ac or dc.

38. Ans (3)

Maxwell's equations are as follows

(i) $\oint \vec{E}.d\vec{A} = \frac{q}{\varepsilon_0}$ (Gauss's law of electricity)

(ii) $\oint \vec{B}.d\vec{A} = 0$ (Gauss's law of magnetism)

(iii) $\oint \vec{E}.d\vec{\ell} = -\frac{d\phi_B}{dt}$ (Faraday's law)

(iv) $\oint \vec{B}.d\vec{\ell} = \mu_0 I_C + \mu_0\varepsilon_0\frac{d\phi_E}{dt}$ (Ampere – Maxwell law)

39. Ans (4)

Electromagnetic waves consists of periodically oscillating electric and magnetic vectors in mutually perpendicular planes but vibrating in phase.

40. Ans (3)

β-rays are electrons

41. Ans (4)

The direction of propagation of electromagnetic wave is perpendicular to both electric field vector $\vec{E}$ and magnetic field vector $\vec{B}$ i.e., in the direction of $\vec{E} \times \vec{B}$.

42. Ans (2)

(A) $I = \mu \times c = \frac{B_0^2}{2\mu_0}c = \frac{cB^2}{\mu_0}$

(B) $\mu = \frac{B_0^2}{2\mu_0}$

(C) $P_r = \frac{EB}{\mu_0 c}$

(D) $P_a = \frac{EB}{2\mu_0 c}$

43. Ans (1)

Compare the given equation with $E = E_0\cos(kz - \omega t)$

We get, $\omega = 6 \times 10^8\ s^{-1}$

$\therefore$ Wave vector, $k = \frac{\omega}{c} = \frac{6 \times 10^8 s^{-1}}{3 \times 10^8 ms^{-1}} = 2\ m^{-1}$

44. Ans (1)

Speed of light, $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$; Also, $\frac{E}{B} = c$

Therefore, $\sqrt{\mu_0\varepsilon_0}\frac{E}{B}$ is dimensionless.

45. Ans (3)

The Quality factor (Q) measures the sharpness of resonance. It is defined as the ratio of the resonant frequency (f_0) to the bandwidth (Δf), which is the difference between the upper and lower half-power frequencies.

Calculation: • Given: $f_0 = 600$ Hz, $f_1 = 550$ Hz, $f_2 = 650$ Hz.

• Bandwidth: $\Delta f = f_2 - f_1 = 650 - 550 = 100$ Hz.

• Q – factor: $Q = \frac{f_0}{\Delta f} = \frac{600}{100} = 6.$

46. Ans (4)

EAN = Z−0.5 + 2 × C.No.

1. $[Fe(CN)6]^{3-}$: Fe Z=26, ox.state = +3, ligands 6×2 =12 ⟶ EAN = 26−3+12 = 35.

$[Fe(NH3)6]^{3+}$: Fe Z=26, ox.state = +3, ligands 6×2 =12 ⟶ EAN = 26−3+12 = 35. (same)

2. $[Cr(NH3)6]^{3+}$: Cr Z=24, ox.state = +3, ligands 6×2 =12 ⟶ EAN = 24−3+12 = 33.

$[Cr(CN)6]^{3-}$: Cr Z=24, ox.state = +3, ligands 6×2 =12 ⟶ EAN = 24−3+12 = 33. (same)

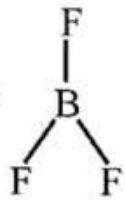
3. $[FeF6]^{3-}$: Fe Z=26, ox.state = +3, ligands 6×2 =12 ⟶ EAN = 26−3+12 = 35.

$[Fe(CN)6]^{3-}$: Fe Z=26, ox.state = +3, ligands 6×2 =12 ⟶ EAN = 26−3+12 = 35. (same)

4. $[Ni(CO)4]$: Ni Z=28, ox.state = 0, ligands 4×2 =8 ⟶ EAN = 28−0+8 = 36.

$[Ni(CN)4]^{2-}$: Ni Z=28, ox.state = +2, ligands 4×2 =8 ⟶ EAN = 28−2+8 = 34. (different)

47. Ans (4)

A : BF_3 :  (sp^2, T.P.)

R : $BF_3 + H_2O \longrightarrow H_3BO_3 + HF$

Scanned with OKEN Scanner

48. Ans (4)

(a) $Na_2[Fe(CN)_5NO] \xrightarrow[Na_2S]{S^{-2}} Na_4[Fe(CN)_5(NOS)]$

(c) $Fe^{+3} + K_4[Fe(CN)_6] \longrightarrow Fe_4[Fe(CN)_6]_3$

(Prussian blue)

49. Ans (3)

(1) $[Cu(NH_3)_4]\,[PtCl_4]$

(2) $[Cu(NH_3)_3Cl]\,[PtCl_3(NH_3)]$

(3) $[Cu(NH_3)Cl_3]\,[PtCl(NH_3)_3]$

(4) $[CuCl_4]\,[Pt(NH_3)_4]$

50. Ans (1)

Assertion : CO_2 is a linear molecule but SiO_2 is tetrahedral.

Reason : $2p\pi$-$3p\pi$ is a weak bond.

51. Ans (4)

(1) $Sc^{+3} = 3d^0$ colourless

$Ti^{+3} \longrightarrow 3d^1$ Violet colour

(2) $[Ma_4b_2] \longrightarrow$ Geometrical isomers

(3) $[Pd(NO_2)_2(NH_3)_2] \longrightarrow$ Square planar

Show geometrical isomers & linkage isomers

52. Ans (4)

$[Co(en)_2Cl_2]^{\oplus}$ → Cis-active

→ Trans optically inactive → have plane of symmetry

53. Ans (3)

$KFe[Fe(CN)_6]$ → $\overset{+1\ +2\ +3}{K\ Fe\ [Fe(CN)_6]}$ → Turn blue

$KFe[Fe(CN)_6]$ → $\overset{+1\ +3\ +3}{K\ Fe\ [Fe(CN)_6]}$ → Prussian blue

54. Ans (4)

$XeF_6 + SbF_5 \longrightarrow \underset{(sp^3d^2)\ \text{square planar}}{XeF_5^{+}} + \underset{(sp^3d^2)\ \text{Octahedral}}{SbF_6^{\ominus}}$

55. Ans (1)

d^4 → eg^1, t_{2g}^3

CFSE = $[-1.2 + 0.6]\ \Delta_0 = -0.6\ \Delta_0$

$\Rightarrow [Fe(\eta^5 - C_5H_5)_2]$ C.N. of Fe = 6

56. Ans (1)

In [M abcdef] POS abscent

57. Ans (2)

In Electro plating = $Na[Ag(CN)_2]/Na[Au(CN)_2]$

58. Ans (1)

$Pb^{2+}(aq) + 2Cl^{-}(aq) \longrightarrow PbCl_2(s)$

$Pb^{2+}(aq) + H_2S(g) \longrightarrow PbS(s) + 2H^{+}(aq)$

Black

$Fe^{3+}(aq) + 3OH^{-}(aq) \longrightarrow Fe(OH)_3(s)$

A reddish-brown precipitate forms.When Ni^{2+} (nickel(II) ions) react with dimethylglyoxime (DMG) solution, a characteristic bright red precipitate of nickel dimethylglyoxime complex forms.

$Zn^{2+}(aq) + 4CN^{-}(aq) \longrightarrow [Zn(CN)_4]^{2-}$

$[Zn(CN)_4]^{2-}$, which is colorless and soluble in water.

59. Ans (1)

MnF_6 does not exist.

60. Ans (1)

$NH_4Cl \longrightarrow \boxed{NH_4^{+}} + Cl^{-}$

$NH_4OH \longrightarrow \boxed{NH_4^{+}} + OH^{-}$

(WB) Common ion

degree of dissociation of NH_4OH decreases. So, concentration of OH^{-} decreases.

61. Ans (4)

(1) Be^{+2} and Mg^{+2} do not impact colour in flame test due to their High I.P.

(2) H.E. : $Be^{+2} > Mg^{+2}$

$\left[HE \propto \frac{1}{r_+ + r_-}\right]$

62. Ans (1)

(1) $K_2Cr_2O_7 + CO_2 \longrightarrow$ No reaction

CO_2 does not act as R.A

(2) $K_2Cr_2O_7 + SO_2 \longrightarrow \underset{\text{Green}}{Cr^{+3}} + SO_3$

Scanned with OKEN Scanner

63. Ans (1)

(1) $Fe^{+3}, A\ell^{+3}, Cr^{+3} - NH_4OH + NH_4Cl$

(2) $(NH_4)_2CO_3 + NH_4OH - Ba^{+2}, Sr^{+2}, Ca^{+2}$

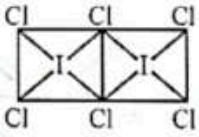
(3) $NH_4OH + H_2S - Mn^{+2}, Zn^{+2}$

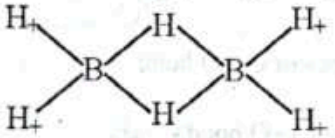
(4) Dil HCl $- Pb^{+2}$

64. Ans (4)

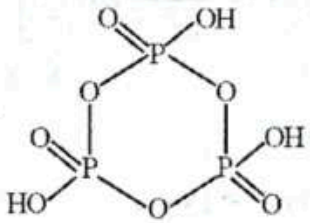
$Zn^{+2} + H_2S \xrightarrow{NH_4OH} ZnS$ (white ppt)

65. Ans (4)

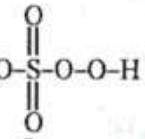
(1) I_2Cl_6 - Planar 

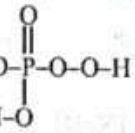
(2) 

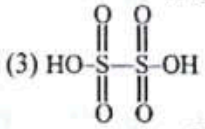
66. Ans (2)

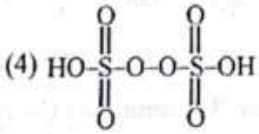
$(HPO_3)_3 \rightarrow$ 

67. Ans (3)

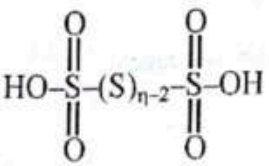
(1) H_2SO_5 : 

(2) H_3PO_5 : 

(3) 

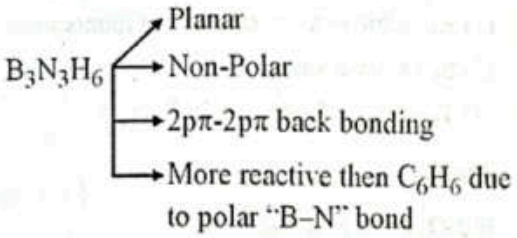
(4) 

68. Ans (3)



Total no. of 'S' atom in zero O.S. = (n – 2)

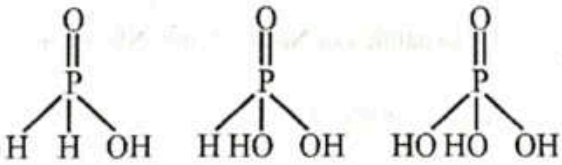
69. Ans (2)



70. Ans (4)

Due to small size of Li^+ it can't form any type of Alum.

71. Ans (4)



→ No. of P–H ↓es = Reducing power ↓es

→ Basicity ↑es

$H_3PO_3 \xrightarrow{\Delta} PH_3 + H_3PO_4$

$H_4P_2O_7 \rightarrow$ No P–H bond

72. Ans (3)

	F	Cl	Br	I	At
EN	4	3	2.8	2.5	2.2

73. Ans (1)

Fe + Conc. HCl $\rightarrow FeCl_2 + H_2\uparrow$

$FeCl_3$ not formed, H_2 gas prevent the formation of $FeCl_3$

74. Ans (3)

C_2H_6 is a dimer of 'CH_3' (odd e^- species)

C_2H_6 form by 2C–2e

75. Ans (1)

$Na_2B_4O_7.10H_2O + Cr^{+3} \xrightarrow{\Delta} Cr(BO_2)_3$

(Green)

Scanned with OKEN Scanner

76. Ans (1)

(1) Rhombic is more stable than monoclinic

(2) S_8 - Crown shape

(3) Total No. of lone pair in $S_8 = 16$

77. Ans (4)

H_3BO_3 not a protonic acid

H_3BO_3 is a weak Lewis acid

H_3BO_3 Mono basic acid

78. Ans (1)

(1) In halide's of Nitrogen only NF_3 stable.

(2) BiF_5 is stable

(3) Nitrogen do not show allotropy

(4) $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$ = order of stability

79. Ans (3)

Assertion : $K_2Cr_2O_7 + C_2H_5OH \xrightarrow{H^+} Cr^{+3} + CH_3COOH$

Reason : $Na_2Cr_2O_7$ show Hygroscopic nature.

80. Ans (4)

Tri Halides of group-15 are stable except Nitrogen as (NCl_3, NBr_3 and NI_3) are not stable because of steric hindrance or steric repulsion. (small size of central atom and large size of side atoms cause Crowding which decreases Sterility)

81. Ans (4)

$$E^{o}_{Co^{+3}/Co^{+2}} \rangle E^{o}_{Mn^{+3}/Mn^{+2}} \rangle E^{o}_{Mn^{+3}/Mn^{+2}}$$

82. Ans (3)

$K_2Cr_2O_7 + H_2O_2 \xrightarrow{H^+} CrO_5$

83. Ans (2)

$KMnO_4 + HCl \rightarrow Mn^{+2} + Cl_2\uparrow$

O.A R.A

84. Ans (2)

(a) $HOCl > HClO_2 > HClO_3 > HClO_4$ - Order of oxidising power

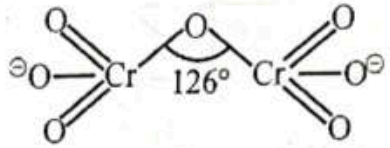
$\rightarrow$ T.S. $\uparrow$es = oxidising power $\downarrow$es

+4 +4 +4

(b) GeO_2 SnO_2 PbO_2

$\rightarrow$ Stability of HOS $\downarrow$es = oxidising power $\uparrow$es

85. Ans (4)



(1) Two types of Cr–O bond

(2) Identical Cr–O bond's = six

$$\left[Cr_2O_7^{-2} \underset{H^{\oplus}}{\overset{\Delta H^{\ominus}}{\rightleftharpoons}} CrO_4^{-2}\right]$$

86. Ans (4)

(1) $Yb^{+2} \Rightarrow [Xe]4f^{14}$

$Eu^{+2} \Rightarrow [Xe]4f^{7}$

(2) $Ce^{+4} \Rightarrow [Xe]$

$Tb^{+4} \Rightarrow [Xe]4f^{7}$

(3) $Gd \xrightarrow{IP_1} Gd^{+} \xrightarrow{IP_2} Gd^{+2} \xrightarrow{IP_3} Gd^{+3}$

$[Xe]4F^{7}5d^{1}$ $[Xe]4F^{7}$

87. Ans (4)

$FeCr_2O_4 \xrightarrow{Na_2CO_3+O_2} Fe_2O_3 + Na_2CrO_4 + CO_2$

88. Ans (1)

(1) Reactivity of earlier "Ln" similar to Ca and Higher "Ln" similar "Al"

(2) Ce^{+4} is good Analytical reagent.

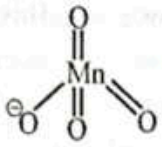
89. Ans (2)

$MnO_2 + 4HCl \xrightarrow{H^+} Mn^{+2} + Cl_2\uparrow + 2H_2O$

Scanned with OKEN Scanner

90. **Ans (3)**

$MnO_4^{\ominus}$ is a Tetrahedral, (Purple)



$[p\pi\text{-}d\pi]$

Due to resonance all Mn–O bond length are same.

91. **Ans (1)**

NCERT-XII, Page-54

92. **Ans (4)**

NCERT-XII, Pg. # 67

93. **Ans (3)**

NCERT-XII, Pg. # 67

94. **Ans (3)**

NCERT Pg. # 60, 61, 62 (E)

67, 68, 69 (H)

95. **Ans (1)**

NCERT XII Pg. # 60

96. **Ans (1)**

NCERT-XII, Pg. # 57, 63

97. **Ans (1)**

NCERT Page No. 72

98. **Ans (3)**

NCERT-XII, Pg # 74

99. **Ans (4)**

NCERT-XII, Pg. # 61

100. **Ans (1)**

NCERT-XII, Pg. # 58, 59

101. **Ans (3)**

NCERT Pg. No. 59, 61, 64, 69

102. **Ans (1)**

NCERT Pg. # 65, 67

103. **Ans (3)**

NCERT-XII, Pg. # 59

104. **Ans (1)**

NCERT (XIIth) Pg. # 59

105. **Ans (2)**

NCERT Pg. # 76

106. **Ans (4)**

NCERT Pg.#58,60

107. **Ans (1)**

NCERT Pg.#58(E), 65(H)

108. **Ans (2)**

NCERT Pg # 67 (E) / 76 (H)

109. **Ans (2)**

NCERT, Pg. # 73

110. **Ans (4)**

NCERT, Pg. # 60,62 / 67,69

111. **Ans (3)**

NCERT XII Pg. No. # 77

112. **Ans (4)**

NCERT, Pg # 90,89,85

113. **Ans (3)**

NCERT Pg. # 85 (E) / 94 (H)

114. **Ans (2)**

NCERT Pg. # 81, 91, 95

115. **Ans (2)**

NCERT-XII, Pg. # 85, 86

116. **Ans (4)**

NCERT Pg. # 93, 94

117. **Ans (2)**

NCERT Pg. # 80-81/89-98

118. **Ans (2)**

NCERT, Pg. # 97

119. **Ans (3)**

NCERT-XII, Pg. # 96

120. **Ans (1)**

NCERT-XII, Pg. # 89

Scanned with OKEN Scanner

121. Ans (2)
NCERT Pg.# English- 87,90.
Hindi- 97,99.

122. Ans (1)
NCERT Pg#83,95

123. Ans (2)
NCERT-XII, Pg. # 81

124. Ans (3)
NCERT Pg. # 83

125. Ans (2)
NCERT Pg. No. # 83, 101, 95

126. Ans (2)
NCERT Pg.#102-103

127. Ans (3)
NCERT Pg#97

128. Ans (2)
NCERT (XII)th Pg. # 96,97,98

129. Ans (2)
NCERT (XII)th Pg. # 97

130. Ans (3)
NCERT Pg. # 81-82/91

131. Ans (3)
NCERT Pg.#95.

132. Ans (3)
NCERT Pg # 98, 99

133. Ans (1)
NCERT-XII, Pg. # 101

134. Ans (3)
NCERT-XII, Pg. # 93, Fig. 6.10

135. Ans (2)
NCERT Page No. # 91

136. Ans (1)
Transgenic mice are genetically modified to express specific human proteins or genes. This makes them valuable models for studying human diseases and testing the safety and efficacy of drugs and vaccines. In the case of polio vaccines, transgenic mice can be engineered to express human poliovirus receptors, allowing researchers to test vaccine efficacy and safety in a controlled laboratory setting.

137. Ans (1)
The first transgenic cow, Rosie, produced human protein-enriched milk (α-lactalbumin).
This was a significant achievement in biotechnology as it demonstrated the potential of transgenic animals to produce valuable human proteins in their milk. The a-lactalbumin protein is important for infant nutrition, and Rosie's milk was enriched with this protein, making it a valuable resource for human health.

138. Ans (2)
NCERT Pg. # 196, 197

139. Ans (2)
NCERT (XII) ; Page No. # 179, 180, 184

140. Ans (1)
NCERT-XII, Pg # 179

141. Ans (3)
NCERT-XII, Pg # 180,186

142. Ans (1)
Question Explanation : Asking the techniques, which serves for the purpose of early diagnosis of a disease

Concept : Tools and Techniques

Solution :The techniques used for the purpose of early diagnosis are -
(1) PCR (2) RDT (3) ELISA

Final Answer : 1

Scanned with OKEN Scanner

143. **Ans (1)**
NCERT, Pg. # 184

144. **Ans (2)**
NCERT 12th, Page No. # 181,182,183 (E)/ 199, 200 (H)

145. **Ans (4)**
NCERT Pg. # 4

146. **Ans (2)**
NCERT, Pg. # 182

147. **Ans (4)**
NCERT-XII, Page # 180

148. **Ans (4)**
The first clinical gene therapy was conducted in 1990 on a 4-year-old girl name Ashanthi DeSilva, who had ADA (Adenosine Deaminase) deficiency, a genetic disorder affecting the immune system. The therapy involved inserting a functional gene to correct the deficiency.

149. **Ans (1)**
The animal being used to test the safety of the polio vaccine is Transgenic mice.
Mice are commonly used in scientific research, including vaccine development, due to their rapid reproduction rate, short lifespan, and genetic similarity to humans. By genetically modifying mice to express human proteins or genes involved in the poliovirus infection process, researchers can study the virus's behavior and test the effectiveness and safety of vaccines in a controlled environment.

150. **Ans (3)**
NCERT-XII, Pg. # 184, 185 (E), 202, 203 (H)

151. **Ans (1)**
NCERT Pg. No. # 178

152. **Ans (4)**
New NCERT Pg. No. 179, 180

153. **Ans (3)**
NCERT-XII, Pg. # 184(E), 201, 202(H)

154. **Ans (3)**
NCERT XII, Pg. No. 182

155. **Ans (1)**
NCERT-XII, Pg. # 179

156. **Ans (3)**
NCERT-XII, Pg. # 183, 184

157. **Ans (3)**
They increase nutritional quality of curd

158. **Ans (1)**
The correct answer is **1. Bacteria**.
Explanation: The large holes in Swiss cheese are formed due to the production of carbon dioxide (CO_2) by bacteria, particularly *Propionibacterium sharmanii*, during the fermentation process. These bacteria convert lactic acid into CO_2, which creates gas bubbles and results in the characteristic holes in the cheese.

159. **Ans (3)**
Explain Question: Question asks about the organism that is also known as brewer's yeast.
Concept: Microbes in industrial products
Solution: Among the given options, Saccharomyces cerevisiae is the correct organism referred to as brewer's yeast.
Final answer: 3

160. **Ans (1)**
NCERT XII Pg # 157

161. **Ans (4)**
NCERT Pg. # 151,152,157,158

162. **Ans (1)**
NCERT (XII) ; Page No. # 152,153

163. **Ans (3)**
NCERT (XII) ; Page No. # 168

164. **Ans (1)**
NCERT, Pg. # 153, 154

Scanned with OKEN Scanner

165. **Ans (2)**
NCERT Pg. # 153

166. **Ans (4)**
NCERT, Pg # 180 to 183

167. **Ans (1)**
NCERT, Pg # 179,184,168,167

168. **Ans (2)**
NCERT, Pg # 173 to 175, 183 & 184

169. **Ans (4)**
NCERT, Pg # 183

170. **Ans (4)**
NCERT, Pg # 179

171. **Ans (1)**
NCERT-XII Pg#181(E),198(H)

172. **Ans (2)**
NCERT-XII, Pg. # 178

173. **Ans (1)**
NCERT-XII, Pg. # 178

174. **Ans (4)**
NCERT (XII) Page No. 184, 185

175. **Ans (4)**
NCERT Pg. No. # 180

176. **Ans (1)**
NCERT-XII, Pg. # 181 (E), 198 (H)

177. **Ans (3)**
NCERT-XII, Pg. # 182 (E), 199, 200 (H)

178. **Ans (2)**
NCERT Pg. # 151, 152

179. **Ans (2)**
NCERT Pg. # 154-158

180. **Ans (2)**
NCERT Pg.#153

Scanned with OKEN Scanner